UNIVERSIDAD
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OPTIMIZATION

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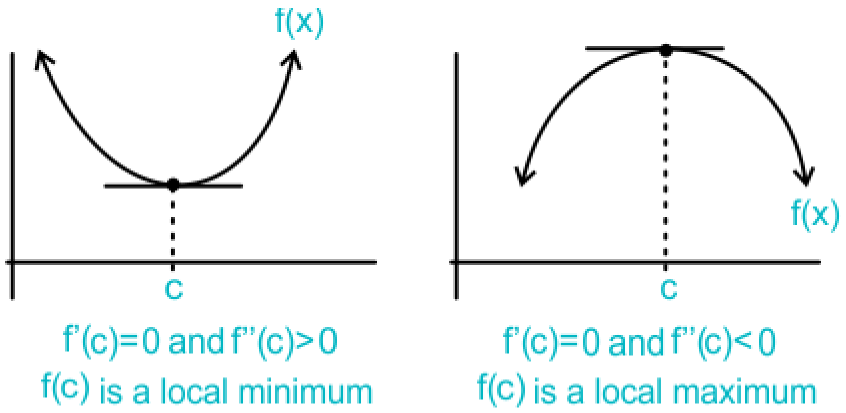
CONTENIDO

1. Introduction

Relationship to Root Finding

- Optimization seeks the points where a function's derivative f′(x) equals zero and uses the second derivative f″(x) to distinguish minima (f″(x)>0) from maxima (f″(x)<0).

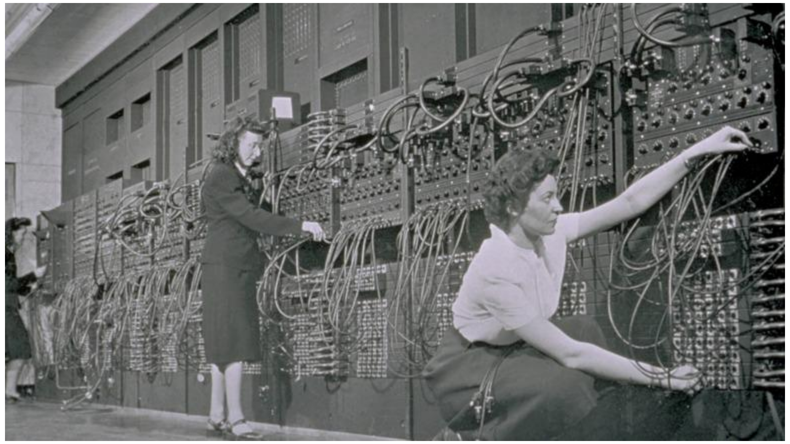
- Many methods reduce optimization to solving the root problem f′(x)=0, often approximating derivatives numerically.



Historical and Non-Computer Methods

- Foundations laid by Bernoulli, Euler and Lagrange (calculus of variations and Lagrange multipliers for constrained problems).

- Post–World War II: Dantzig's simplex method for linear programming, followed by Charnes and colleagues, and rapid growth of unconstrained methods with computer availability.



Optimization in Engineering Practice

- Unlike descriptive (simulation) models, optimization models are **prescriptive**, prescribing the best design or course of action under physical and cost constraints.

- Engineers routinely balance performance, physical limits and cost.

Common Engineering Optimization Examples

- Design aircraft for minimum weight and maximum strength.
- Optimal trajectories of space vehicles.
- Design civil engineering structures for minimum cost.
- Design water-resource projects like dams to mitigate flood damage while yielding maximum hydropower.
- Predict structural behavior by minimizing potential energy.
- Material-cutting strategy for minimum cost.
- Design pump and heat transfer equipment for maximum efficiency.
- Maximize power output of electrical networks and machinery while minimizing heat generation.
- Shortest route of salesperson visiting various cities during one sales trip.
- Optimal planning and scheduling.
- Statistical analysis and models with minimum error.
- Optimal pipeline networks.
- Inventory control.
- Maintenance planning to minimize cost.
- Minimize waiting and idling times.
- Design waste treatment systems to meet water-quality standards at least cost.

Classification of Optimization Problems

By form of f(x):

- Linear programming (linear f and constraints)
- Quadratic programming (quadratic f, linear constraints)
- Nonlinear programming (nonlinear f or constraints)

By constraints

- Unconstrained
- Constrained (with equality/inequality conditions)

By dimensionality

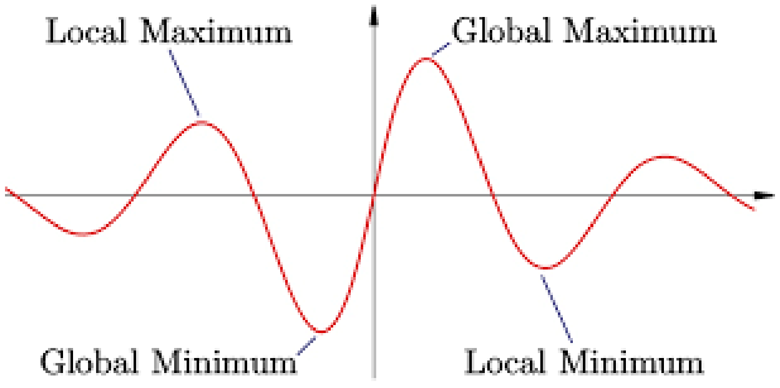
- One-dimensional problems (single variable)
- Multidimensional problems (two or more variables)

2. One-Dimensional Unconstrained Optimization

Overview of One-Dimensional Unconstrained Optimization

- Goal: find the minimum or maximum of a single-variable function f(x).

- Challenges: functions can be **multimodal**, with multiple local extrema; distinguishing the **global** extremum is nontrivial.

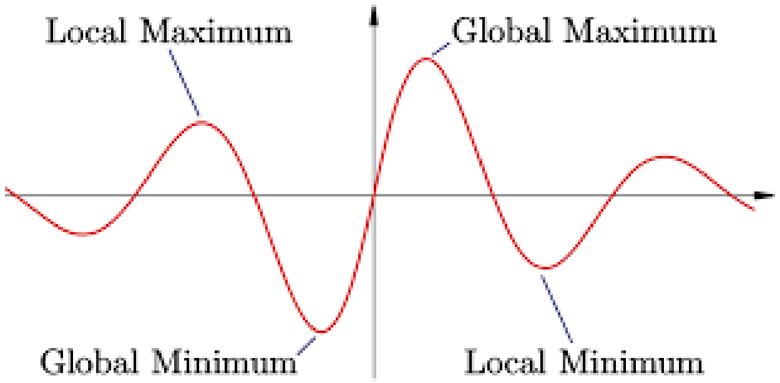


General Strategies for Locating a Global Extremum

1. **Graphical Insight** – visualize low-dimensional functions to identify likely global optima.

2. **Multiple Starting Guesses** – run a local optimizer from varied (possibly random) starting points and select the best result.

3. **Perturbation** – after converging to a local extremum, slightly perturb the solution and rerun to check for better optima.



Classification of One-Dimensional Optimization Methods

- **Bracketing Methods** (require an initial interval [xℓ,xu] containing a single extremum):
  - *Golden-Section Search*
  - *Parabolic Interpolation* (faster convergence but risk of divergence)

- **Open (Point) Methods** (rely on derivatives):
  - *Newton’s Method* (solve f’(x)=0f'(x)=0f’(x)=0 as a root-finding problem)

- **Hybrid Methods**:
  - *Brent’s Method* (combines golden-section reliability with parabolic speed)

Golden-Section Search Key Concepts

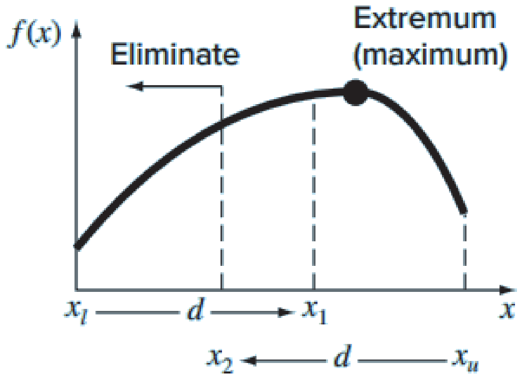
1. **Unimodal Interval** – assume [xℓ,xu]
2. **Interior Points** – choose

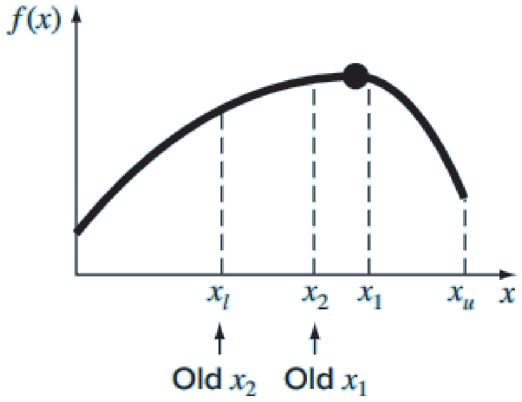
$$d = \frac{\sqrt{5}-1}{2}(x_u - x_l), \qquad x_1 = x_l + d, \qquad x_2 = x_u - d$$

3. Interval Reduction – evaluate f(x_1) and f(x_2):

- If f(x_1) >f(x_2) (seeking maximum), discard [x_l, x_2].
- Otherwise, discard [x_1, x_u].

4. Function-Evaluation Savings – by reusing the interior point from the previous iteration, only one new evaluation per step is needed.





Example 1

Use the golden-section search to find the maximum of

$$f(x) = 2\sin x - \frac{x^2}{10}$$

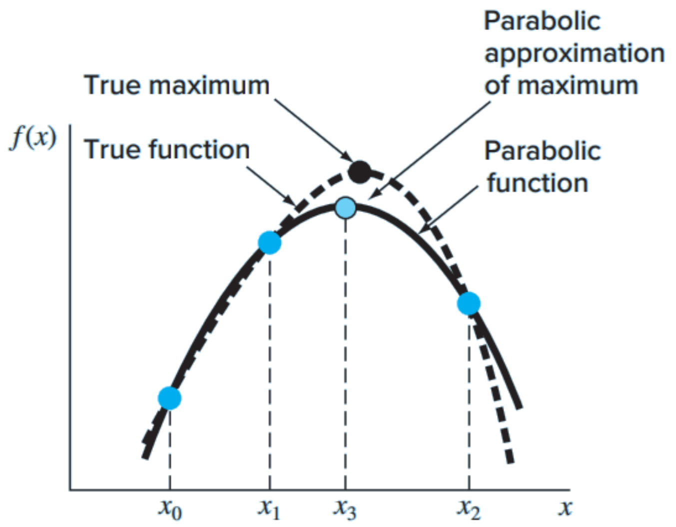
within the interval x_l= 0 and x_u = 4

Parabolic Interpolation

- **Purpose**: Approximate the shape of a function f(x) near its extremum using a quadratic (parabola).
- **Key idea**: Just as a straight line is uniquely defined by two points, a parabola is uniquely defined by three points.
- Formula:

$$x_3 = \frac{f(x_0)(x_1^2 - x_2^2) + f(x_1)(x_2^2 - x_0^2) + f(x_2)(x_0^2 - x_1^2)}{2[f(x_0)(x_1 - x_2) + f(x_1)(x_2 - x_0) + f(x_2)(x_0 - x_1)]}$$

- Where x_0, x_1, x_2 are three guesses bracketing the optimum, and x_3 is the estimated extremum.



Newton’s Method for Optimization

- **Background**: Adaptation of Newton–Raphson root-finding to extrema of f(x).
- **Iteration formula**:

$$x_{i+1} = x_i - \frac{f'(x_i)}{f''(x_i)}$$

- **Characteristics**:
  - **Open method**: Does not require initial bracketing of the optimum.
  - **Rapid (quadratic) convergence** when close to the solution.
  - **Potential divergence** if the initial guess is poor or if f’’(xi) is zero/changes sign.
- **Practical advice**: Always check that the second derivative has the desired sign (positive for a minimum, negative for a maximum) to ensure convergence to the intended extremum.

Example 2

Use the parabolic interpolation and Newton's Method to find the maximum of

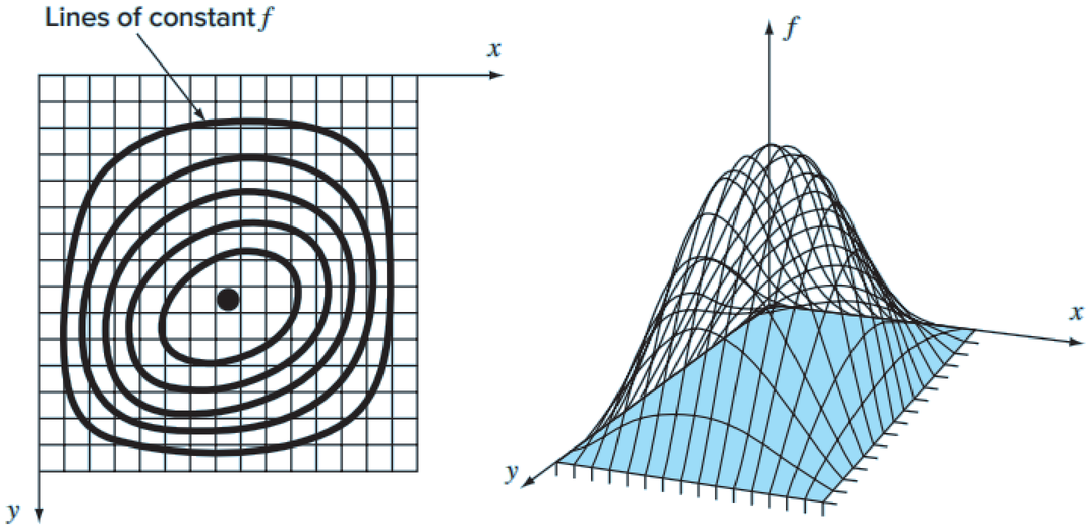
$$f(x) = 2\sin x - \frac{x^2}{10}$$

For parabolic interpolation use an initial guesses of $x_0 = 0$, $x_1 = 1$, and $x_2 = 4$. For Newton's Method use an initial guess of $x_0 = 2.5$

3. Multidimensional Unconstrained Optimization

Overview: Multidimensional Unconstrained Optimization

- Goal: Find minima or maxima of functions of several variables.
- Focus here: Two-dimensional case (mountains and valleys analogy).
- Classification of methods:
  - **Non-gradient (Direct) methods**
  - **Gradient (Descent/Ascent) methods**



Direct Methods (Non-gradient)

Random Search

- Brute-force sampling of function at random points.
- Guarantees finding the global optimum eventually.
- Drawbacks:
  - Highly inefficient as dimensionality grows.
  - Ignores any information about function behavior.

More Sophisticated Heuristics

- Simulated annealing
- Tabu search
- Artificial neural networks
- **Genetic algorithms** (most widely used; pioneered by Holland 1975; overviews by Davis 1991 and Goldberg 1989)

Example 3

Use a random number generator to locate the maximum of

$$f(x, y) = y - x - 2x^2 - 2xy - y^2$$

in the domain bounded by x = −2 to 2 and y = 1 to 3.

4. Constrained Optimization

Constrained Optimization Overview

- Deals with optimization problems where constraints play a role.
- When both the objective function and the constraints are linear, specialized methods (linear programming) can solve very large problems efficiently.
- Applicable in diverse engineering and management settings.
- Nonlinear constrained optimization is discussed more briefly, and software package options are surveyed.

Linear Programming (LP) Definition

- An LP problem seeks to maximize (or minimize) a linear objective (e.g., profit or cost) subject to linear constraints representing limited resources.
- “Linear” refers to the fact that both the objective and all constraints are linear functions of the decision variables.
- “Programming” here means “scheduling” or “setting an agenda,” not computer programming.

Graphical Solution (for Two-Variable LPs)

- Plot each linear constraint as a straight line in the $x_1 - x_2$ plane (with x_1 on the horizontal axis, x_2 on the vertical axis).

- The region where all constraints overlap (including $x_1 \geq 0$ and $x_2 \geq 0$) is the feasible solution space, containing all points that satisfy every constraint.

Example 4

Develop a graphical solution for the following problem:

$$Maximize\ Z = 150x_1 + 175x_2$$

subject to

$7x_1 + 11x_2 \leq 77$

$10x_1 + 8x_2 \leq 80$

$x_1 \leq 9$

$x_2 \leq 6$

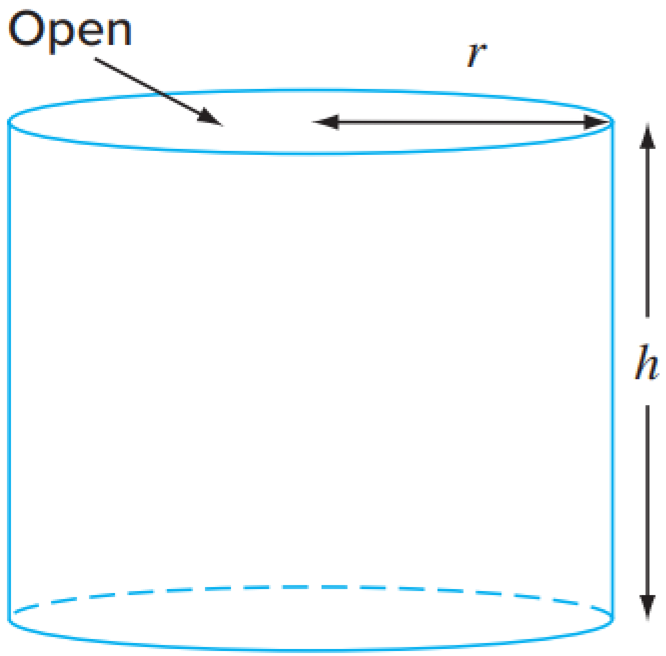
$x_1 \geq 0$

$x_2 \geq 0$

5. Case Studies: Optimization

Example 5

Design the optimal cylindrical container that is open at one end and has walls of negligible thickness. The container is to hold $0.2\ m^3$. Design it so that the areas of its bottom and sides are minimized.

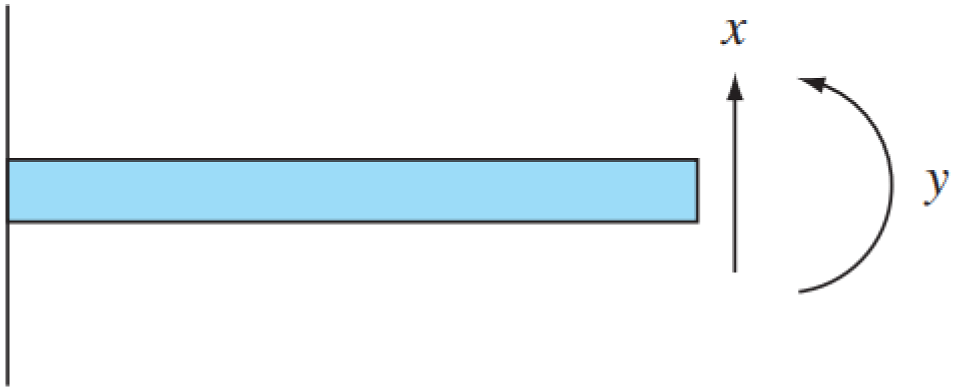


Example 6

A finite-element model of a cantilever beam subject to loading and moments is given by optimizing

$$f(x, y) = 5x^2 - 5xy + 2.5y^2 - x - 1.5y$$

where x = end displacement and y = end moment. Find the values of x and y that minimize f(x, y).

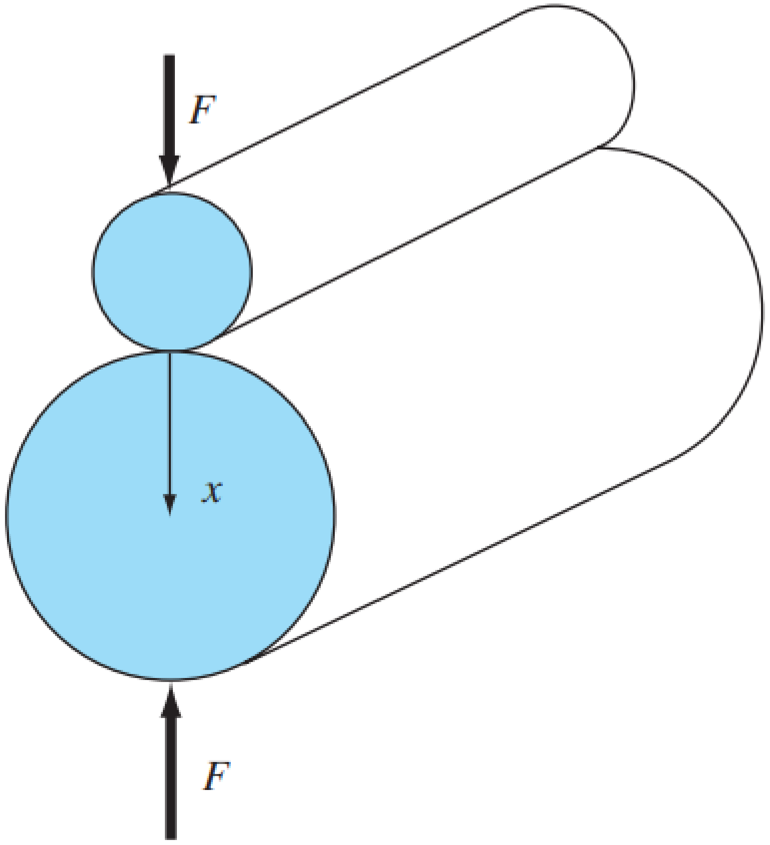


Example 7

Roller bearings are subject to fatigue failure caused by large contact loads F. The problem of finding the location of the maximum stress along the x axis can be shown to be equivalent to maximizing the function

$$f(x) = \frac{0.4}{\sqrt{1+x^2}} - \sqrt{1+x^2}\left(1 - \frac{0.4}{1+x^2}\right) + x$$

Find the value of x that maximizes f(x) with Golden-Section Search.



6. Summary

Summary

- **Definition of Optimization**
  Finding points where a function's derivative $f'(x)=0$ and using $f''(x)$ to distinguish minima ($f''>0$) from maxima ($f''<0$); many algorithms reduce to solving root-finding problems and approximate derivatives numerically .

- **Historical Foundations**
  Early work by Bernoulli, Euler, and Lagrange (calculus of variations, Lagrange multipliers); post–WWII advances include Dantzig's simplex method for linear programming and rapid growth of unconstrained methods with the advent of computers .

- **Role in Engineering Practice**
  Optimization models are prescriptive (unlike descriptive simulation), prescribing best designs or actions under physical and cost constraints; engineers balance performance, physical limits, and cost .

Summary

- **One-Dimensional Unconstrained Optimization**
  - **Goal:** Find global minima/maxima of single-variable functions, which may be multimodal.
  - **Methods:**
    Bracketing: Golden-section search; parabolic interpolation
    Open (derivative-based): Newton’s method

- **Multidimensional Unconstrained Optimization**
  Direct (non-gradient) methods: Random search

- **Constrained Optimization**
  - **Linear Programming (LP):** Maximize/minimize a linear objective subject to linear constraints (“programming” meaning scheduling)
  - **Solution approaches:** Simplex method; graphical solutions for two variables